



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

AMERICAN INSTRUMENT CORPORATION

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CALIBRATION

Valid To: December 31, 2018

Certificate Number: 1354.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Chemical

Parameter/Equipment	Range	CMC ² (±)	Comments
pH Meters & Sensors ³	(4, 7, 10) pH	0.023 pH	MV & temperature, calibration standard buffers

II. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ² (±)	Comments
DC Voltage ³ – Generate	(0 to 100) mV (0 to 1) V (0 to 10) V	0.002 mV 0.0076 mV 0.076 mV	Martel M3001
DC Voltage ³ – Measure	(0 to 75) mV (0 to 10) V	0.058 mV 0.25 mV	Martel M3001
DC Current ³ – Generate	(0 to 100) mA	0.0048 mA	Martel M3001

Parameter/Equipment	Range	CMC ² (±)	Comments
DC Current ³ – Measure	(0 to 50) mA	0.0031 mA	Martel M3001
Resistance ³ – Generate	(5 to 400) Ω (5 to 4) kΩ	0.017 Ω 0.35 Ω	Martel M3001
Resistance ³ – Measure	(5 to 400) Ω (5 to 4) kΩ	0.0058 Ω 0.012 Ω	Martel M3001
Electrical Calibration of RTD Indicators ³ – Recorders, Controllers & Calibrators			
Pt 385, 100 Ω	(-200 to 800) °C	0.17 °C	Martel M3001
Pt 3926, 100 Ω	(-200 to 630) °C	0.17 °C	
Pt 3916, 100 Ω	(-200 to 630) °C	0.17 °C	
Pt 385, 200 Ω	(-200 to 630) °C	0.60 °C	
Pt 385, 500 Ω	(-200 to 630) °C	0.27 °C	
Pt 385, 1000 Ω	(-200 to 630) °C	0.19 °C	
Ni 120, 120 Ω	(-80 to 260) °C	0.16 °C	

Parameter/Equipment	Range	CMC ² (±)	Comments
Electrical Calibration of Thermocouple Indicators ³ – Recorders, Controllers, & Calibrators			
Type B	(600 to 800) °C (800 to 1000) °C (1000 to 1550) °C (1550 to 1820) °C	0.78 °C 0.65 °C 0.64 °C 0.70 °C	Martel M3001
Type C	(0 to 150) °C (150 to 650) °C (650 to 1000) °C (1000 to 1800) °C (1800 to 2316) °C	0.56 °C 0.50 °C 0.57 °C 0.86 °C 1.4 °C	
Type E	(-200 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 1000) °C (1000 to 1372) °C	0.88 °C 0.44 °C 0.41 °C 0.43 °C 0.48 °C	
Type J	(-210 to -100) °C (-100 to -30) °C (-30 to 150) °C (150 to 760) °C (760 to 1200) °C	0.48 °C 0.34 °C 0.30 °C 0.32 °C 0.41 °C	
Type K	(-200 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 1000) °C (1000 to 1372) °C	0.56 °C 0.43 °C 0.38 °C 0.51 °C 0.72 °C	
Type N	(-200 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 410) °C (410 to 1300) °C	0.72 °C 0.40 °C 0.33 °C 0.32 °C 0.47 °C	
Type R	(0 to 250) °C (250 to 400) °C (400 to 1000) °C (1000 to 1750) °C	0.95 °C 0.58 °C 0.55 °C 0.67 °C	
Type S	(0 to 250) °C (250 to 1000) °C (1000 to 1400) °C (1400 to 1750) °C	0.85 °C 0.60 °C 0.62 °C 0.76 °C	
Type T	(-250 to -150) °C (-150 to 0) °C (0 to 120) °C (120 to 400) °C	1.1 °C 0.53 °C 0.31 °C 0.28 °C	

III. Mechanical

Parameter/Equipment	Range	CMC ² (±)	Comments
Indirect Verification of Rockwell Hardness Testers & Rockwell Superficial Hardness ³	HRA: Low	0.55 HRA	Indirect verification method per ASTM E18
	Middle	0.54 HRA	
	High	0.44 HRA	
	HRBW: Low	0.77 HRBW	
	Middle	0.84 HRBW	
	High	0.68 HRBW	
	HRC: Low	0.44 HRC	
	Middle	0.51 HRC	
	High	0.45 HRC	
	HREW: Low	0.63 HRE	
	Middle	0.69 HRE	
	High	0.70 HRE	
	HRKW: Low	0.70 HRK	
	Middle	0.75 HRK	
	High	0.79 HRK	
	HR15N: Low	0.38 HR15N	
	Middle	0.44 HR15N	
	High	0.40 HR15N	
	HR30N: Low	0.57 HR30N	
	Middle	0.44 HR30N	
	High	0.62 HR30N	
	HR45N: Low	0.55 HR45N	
	Middle	0.36 HR45N	
	High	0.39 HR45N	
	HR15TW: Low	0.65 HR15T	
	Middle	0.44 HR15T	
	High	0.62 HR15T	
	HR30TW: Low	0.69 HR30T	
	Middle	0.48 HR30T	
	High	0.62 HR30T	

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Indirect Verification of Rockwell Hardness Testers & Rockwell Superficial Hardness ³ (cont)	HR45TW: Low Middle High HR15YW: Low High	0.77 HR45T 0.56 HR45T 0.55 HR45T 1.4 HR15Y 0.89 HR15Y	Indirect verification method per ASTM E18
Indirect Verification of Brinell Hardness Testers at Test Condition(s) ³ – HBW 10/500/15 HBW 10/1500/15 HBW 10/3000/15	(20 to 150) HBW (45 to 700) HBW (45 to 700) HBW	1.5 HBW 6.5 HBW 4.9 HBW	Indirect verification method per ASTM E10
Direct Verification of Brinell Hardness Testers ³ – Verification of the Test Force Verification of the Device for Measuring Indentation Diameters	(500, 1500, 3000) kgf Up to 6 mm	0.21 % 0.0027 mm	Direct verification method per ASTM E10 Verification of the test force is by proving ring per the method of ASTM E4 Stage micrometer
Indirect Verification of Equotip (Leeb) Hardness Testers ³	750 LD	16 LD	ASTM A956

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Indirect Verification of Microindentation Hardness Testers ³ – Knoop Vickers	(25 to 2000) HK (25 to 2000) HV	(3.7 + 0.007 %) HK (2.1 + 0.014 %) HV	Indirect verification method per ASTM E384
Torque Wrenches ³	(5 to 250) in·lbf (10 to 600) ft·lbf	1.4 % 1.3 %	Torque calibrator
Pressure Gauges ³	(0 to 30) psi (30 to 500) psi	0.055 psi 0.15 psi	Pressure calibrator
RPM ³ – Calibration of Tachometer	(0 to 10 000) RPM	(2.6 + 0.0005 %) RPM	Martel MC-1010
Calibration of Photohelic, Magnehelic & Environmental Gauge ³	(0 to 100) in·H ₂ O	0.23 in·H ₂ O	Dwyer 477A manometer
Foundry Sand Test Equipment ³ – Core Scratch Hardness Tester Green Sand Hardness Tester Permmeter Sand Rammer Sand Strength Tester Moisture Teller Friability Tester AFS Clay Tester	Up to 100 in (0 to 100) B & C (0 to 500) perms ----- (0 to 500) psi (0 to 400) °F 60 s (0 to 10) min	0.94 in 0.048 in 0.59 perms 0.010 in 3.8 psi 2.1 °F 1.2 s 1.2 s	Standard test block, flat surface test Flat surface test Perm standard Certified caliper, impact rings Certified load cell Temperature calibrator Certified stop watch Certified stop watch

IV. Thermodynamics

Parameter/Equipment	Range	CMC ^{2, 4, 5} ($\pm$)	Comments
IR Temperature ³	(125 to 900) °F	(2.0 + 0.018 %) °F	Hart 9132 black body
Temperature – Measure ³	(200 to 1000) °C (1000 to 1372) °C	1.2 °C 2.3 °C	Martel PTC-8010 with a type K thermocouple
Liquid-in-Glass Thermometers & Temperature Measuring Devices ³	(0 to 100) °C	0.21 °C	Constant temperature bath, certified sensor
Relative Humidity – Measure	(10 to 95) % RH	2.5 % RH + 0.6R	Rotronics hygro palm

V. Time & Frequency

Parameter/Equipment	Range	CMC ^{2, 5} ($\pm$)	Comments
Frequency – Measure ³	(1.0 to 1000) Hz (1.0 to 10) kHz	(0.12 + 0.0006 %) μ Hz	Martel MC-1010
Frequency – Measuring Equipment ³	(1.0 to 1000) Hz (1.0 to 10) kHz	(0.066 + 0.0005 %) μ Hz	Martel MC-1010

¹ This laboratory offers commercial calibration service and field calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ Field calibration service is available for this calibration and this laboratory meets A2LA R104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

⁴ In the statement of CMC, L is the numerical value of the nominal length of the device measured in inches, and R is the resolution of the device under test.

⁵ In the statement of CMC, percentages are to be read as percent of reading, unless otherwise noted.



Accredited Laboratory

A2LA has accredited

AMERICAN INSTRUMENT CORPORATION

Hartland, WI

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSLI Z540-1-1994 and R205 – *Specific Requirements: Calibration Laboratory Accreditation Program*. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 16th day of January 2017.

A handwritten signature in black ink, appearing to be 'L. S.', written over a horizontal line.

President and CEO
For the Accreditation Council
Certificate Number 1354.01
Valid to December 31, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.